



Computational study of aerodynamic characteristics in flapping MAVs at different angles of attack

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Abstract. In this study, we conducted numerical simulations of a rectangular wing's flapping motion within a three-dimensional flow field using the discrete vortex method (DVM). This method is more computationally efficient because it avoids the need for generating a flow field grid at each time step, unlike the traditional simulation techniques. We evaluated the wing's aerodynamic performance by measuring the lift and drag. The wing was subjected to a combination of chord-wise twisting and span-wise bending at varying angles of attack. In addition, a prescribed motion was imposed on the flapping wing motion. Our findings indicate that the deformable wing produces significantly more lift than a rigid wing, with only a slight increase in drag. Consequently, the lift-to-drag ratio demonstrates that deformable wings are more efficient at generating lift within a moderate angle of attack range, typically between 10° and 20° . A numerical model is employed to capture the basic wake shape and to account for more complex fluidic phenomena, such as vortex stretching. The insights from our numerical simulations are expected to provide a valuable direction for modeling the micro-air vehicles (MAVs).

Keywords. 3D unsteady aerodynamics; rigid wing; deformable wing; lift; drag; angle of attack.

1. Introduction

In contemporary military operations and the management of natural disasters, drones and micro-air vehicles (MAVs) are widely used for aerial surveillance. To create small flapping drones, it is necessary to emulate the flapping flight of insects and birds through biomimicry. In recent times, there has been a lot of research on designing and optimizing MAVs [1–3]. These compact aeroelastic aerial systems are intended to function in urban areas and confined spaces, such as indoor areas, tunnels, and caves. They are expected to carry out a range of missions, including inspecting and observing harsh environments that are unreachable by other types of vehicles or where human life is at risk. As a result, the design must adhere to rigorous performance criteria, encompassing exceptional maneuverability at low speed-flight, strong lift to achieve stable flight, hovering capabilities, and structural resilience to withstand potential impacts and gust loads. Two primary propulsion mechanisms, flapping wings and revolving helicopter blades, can satisfy these requirements [4]. Biomimicry of the flapping motion of birds and insects can

lead to optimum performance of flapping-based small MAVs [5, 6].

Over the past 30 years, there has been increasing interest in the development of small aerial vehicles (UAVs), particularly with the initiation of microtechnologies in the 1990s. These UAVs, also referred to as nanoair vehicles, are typically less than 100 mg in weight and have dimensions of the order of centimeters [7, 8]. Exploring the use of biologically inspired flapping wings as a replacement for traditional fixed and rotating wings for extremely small UAVs has been prompted by their ability to operate at Reynolds numbers below 1000. This is significant because the aerodynamic efficiency of the lifting surfaces deteriorates at these Reynolds numbers due to the impacts of viscous flow separation and friction [9].

Researchers have extensively investigated unsteady flow phenomena, particularly focusing on dynamic stall and methods to suppress it. Dynamic stall occurs when aerofoils undergo time-dependent motion, leading to stall and unsteady separation phenomena, with the effective angle of attack exceeding the angle of stall [10]. This process involves a delayed flow separation/stall onset, accompanied by the vortex releasing from the aerofoil's leading edge, which moves across the aerofoil chord [11]. While previous research on bio-inspired flapping-wing MAVs has extensively explored either flexibility or aerodynamic performance in isolation, a critical gap remains in

understanding the coupled effects of flexible twisting and angle-of-attack dynamics, particularly for bending-twisting coupling (BTC) wings. This study addresses that gap by systematically investigating how controlled wing twist interacts with angle-of-attack variations to enhance lift generation.

The present investigation examines the use of a discrete vortex method to construct a wing flapping in forward flight that resembles that of birds. Vortex methods offer a cost-effective approach to understand complex flows, providing researchers with flexibility and rapid design capabilities [12–15]. They allow gaining a comprehensive grasp of fluid physics in a specific problem, enabling subsequent integration of more advanced experimental and computational methods for higher-fidelity analysis. For many years, these techniques, including panel and particle methods, have been employed in diverse research endeavors and development applications, including the study of flapping wings [16], high angle of pitching plates [17, 18], deforming bodies [10], cyclonic behavior [19], flow around extended spoilers [20], gust response [21], and more.

This study aims to enhance the aerodynamic performance by optimizing the flapping wing's kinematics while adhering to lift and drag constraints. The wing's aerodynamic response is replicated using active shape morphing during forward flight via a three-dimensional (3D) discrete vortex method. The shape of the wing is defined through a combination of twisting and bending, following the approach described in [15, 22–24]. The goal is to develop a computationally efficient numerical code that accurately captures the physical behavior of MAVs. The numerical code is appropriate for Reynolds numbers $Re \sim 10^4 - 10^5$, which are low compared to those in conventional aerodynamics. The study investigates the lift and drag of a deformable flapping wing as the angle of attack varies. The numerical model has captured the wake dynamics, providing a better understanding of the mechanisms of lift and drag. The lift and drag obtained by the wings for different parameters are evaluated to identify the optimum operating regime. The resulting dataset from our numerical simulations will be beneficial for designing MAVs.

2. Computational model

In this study, a discrete vortex method (DVM) was employed to conduct a 3D flapping simulation. The DVM is known for its high accuracy in analyzing unsteady potential flow, which is inviscid, irrotational, and incompressible [12, 25]. At extreme low Reynolds numbers ($Re \sim 1$), such as in bacterial or sperm swimming, viscous effects dominate the creeping flow regime, whereas at high $Re \sim 5 \times 10^5$, viscous dissipation in turbulent flows becomes substantial. Our flow domain ($Re \sim 240 - 1450$) permits an inviscid approximation without significant accuracy loss. As per Katz and Plotkin's book [12], inviscid solutions compare well with viscous flows when the boundary layer remains attached, with only minor lift

reduction due to pressure dampening; this effect intensifies at higher angles of attack ($>50^\circ$), where boundary layer thickening causes trailing-edge separation. However, in flapping wings – where angle-of-attack influence is limited – separation is delayed, justifying the inviscid assumption with minimal deviation from real-case behavior. The wing was divided into a large number of quadrilateral flat surface panels, resulting in $N * M$ panels along the chord and span, respectively (figure 1).

Vortex ring elements were employed to mimic the bound circulation of the wing and vortex wake. Each panel featured a vortex ring positioned at one-quarter of its length and a collocation point situated at the center of the remaining three-quarters of the panel. This collocation point acted at the vortex core, ensuring the no-normal-flow condition. At this location, the normal vector $\mathbf{n}$ was established, and the right-hand rule was used to determine a positive circulation Γ . The vortex ring formulation (eq. 3) was then applied to compute the three velocity components (u, v, w) at any specified location (x, y, z).

2.1 Applying the discrete vortex method

The primary equation of our simulation methodology is the vorticity transport equation:

$$\frac{\partial \Omega}{\partial t} + \mathbf{u} \cdot \nabla \Omega = \Omega(\nabla \cdot \mathbf{u}) + \nu \nabla^2 \Omega. \quad (1)$$

Equation (1) represents velocity as $\mathbf{u}(x, t)$ and vorticity as $\Omega(x, t) = \nabla \times \mathbf{u}(x, t)$. Unlike the 2D flow [26], the 3D flow involves additional complexity due to vortex stretching and tilting. Thus, the resulting modified equation is

$$\frac{D\Omega}{Dt} = \Omega(\nabla \cdot \mathbf{u}). \quad (2)$$

Equation (2) defines the material derivative of vorticity as $\frac{D}{Dt}$. A Lagrangian approach is employed to track the wakes generated by the flapping motion.

The Biot–Savart law expresses the three-dimensional velocity using blob discretization as follows:

$$d\mathbf{v} = \frac{\Gamma}{4\pi} \frac{d\mathbf{l} \times \mathbf{r}}{|\mathbf{r}|^3}. \quad (3)$$

The induced velocity $d\mathbf{v}$ generated from a vortex segment of length $d\mathbf{l}$ and is influenced by both the circulation strength Γ and the distance $\mathbf{r}$ from the segment to the location of interest.

The influence coefficient a_{KL} is the induced normal velocity at the collocation point resulting from a vortex panel with unit strength. For a vortex element L with unit strength and a K th collocation point, the influence coefficient is given by

$$a_{KL} = (\mathbf{u}, \mathbf{v}, \mathbf{w})_{KL} \cdot \mathbf{n}_K. \quad (4)$$

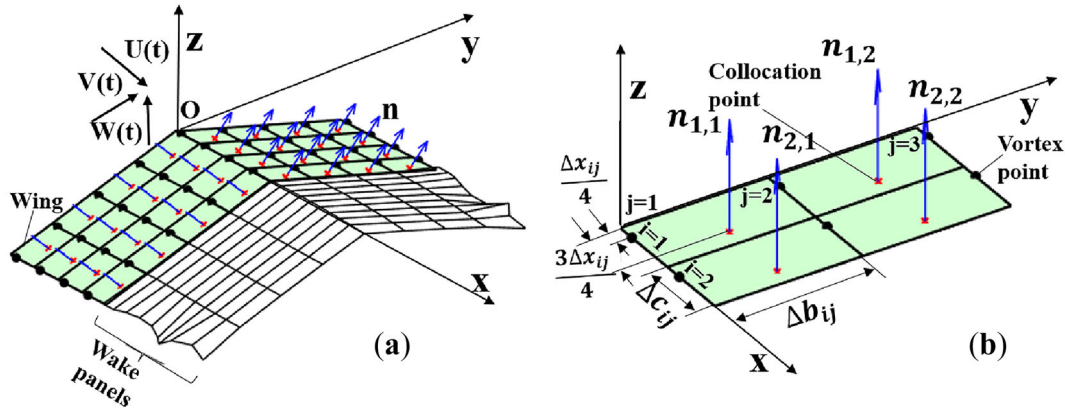


Figure 1. (a) Diagram showing the wing and wake panel and (b) diagram depicting the vortex and collocation point.

Here, L and K represent the horizontal and vertical matrix indices, respectively, and the matrix order is $m = M \times N$.

The equation on the right side pertains to the induced perpendicular flow velocity resulting from the wing motion and previously shed wakes:

$$\text{RHS}_K = -[U(t) + u_w, V(t) + v_w, W(t) + w_w]_K \cdot \mathbf{n}_K. \quad (5)$$

Complete wake causes induced velocity at the K th panel; and U , V , and W are the wing kinematic velocities.

The matrix equation's outcomes can be expressed in the index notation (for every collocation point K) as follows:

$$\sum_{L=1}^m a_{KL} \Gamma_L = \text{RHS}_K. \quad (6)$$

The pressure distribution, lift, and drag coefficients of the wing are calculated by utilizing the unsteady Bernoulli equation:

$$\frac{p_{\text{ref}} - p}{\rho} = \frac{Q^2}{2} - \frac{v_{\text{ref}}^2}{2} + \frac{\partial \Phi}{\partial t}. \quad (7)$$

The subscript ref signifies reference conditions in the distant field, while Q denotes the local fluid velocity, which includes both induced and kinematic velocities.

2.2 Aerodynamic forces

Katz and Plotkin's book on DVM methods [12] states that the pressure disparity among the top and bottom surfaces of an airfoil can be expressed as follows:

$$\begin{aligned} \Delta p_{ij} = \rho \left[(U(t) + u_w, V(t) + u_w, W(t) + w_w)_{ij} \right. \\ \times \tau_i \frac{\Gamma_{i,j} - \Gamma_{i-1,j}}{\Delta c_{ij}} \\ + (U(t) + u_w, V(t) + u_w, W(t) + w_w)_{ij} \\ \left. \times \tau_j \frac{\Gamma_{i,j} - \Gamma_{i,j-1}}{\Delta b_{ij}} + \frac{\partial}{\partial t} \Gamma_{i,j} \right], \end{aligned} \quad (8)$$

where Δc and Δb signify the panel dimension along the chord and span length, respectively, while τ_i and τ_j refer to the panel's tangential vectors along the i and j directions.

The total lift is expressed as

$$L = \sum_{j=1}^N \sum_{i=1}^M \Delta L_{ij} = \sum_{j=1}^N \sum_{i=1}^M \Delta p_{ij} S_{ij} \cos \alpha_{ij}. \quad (9)$$

The induced drag of a three-dimensional lifting surface is specified as

$$\begin{aligned} D &= \sum_{j=1}^N \sum_{i=1}^M [\Delta D_{ij}] \\ &= \sum_{j=1}^N \sum_{i=1}^M \rho \left[(w_{\text{ind}} + w_w)_{ij} \Gamma_{ij} \Delta b_{ij} + \frac{\partial (\Gamma_{ij})}{\partial t} \Delta S_{ij} \sin \alpha_{ij} \right]. \end{aligned} \quad (10)$$

The induced drag of a 3D lifting surface is determined by the downwash wake-induced (w_{ind}) at three quarters of every panel of the wing, panel surface area (ΔS), angle of attack (α), and air density (ρ).

3. Wing geometry and kinematics

Figure 2(a) depicts the wing positioned at a particular angle of attack. In this context, the angle of attack is the angle formed at the root of the leading edge relative to the free stream velocity. Figure 2(b) illustrates the configuration of a deformable wing resulting from a combination of bending and twisting. The objective of incorporating deformation is to explore its potential to improve the wing's flight performance relative to a rigid wing. The deformation of the wing varies sinusoidally throughout the downstroke and reverses during the upstroke. During the flapping motion, the wing twists upwards and bends downwards, generating a substantial positive angle of attack.

Additionally, the twisting angle θ also varies along the span length to maintain lift during flight by ensuring a

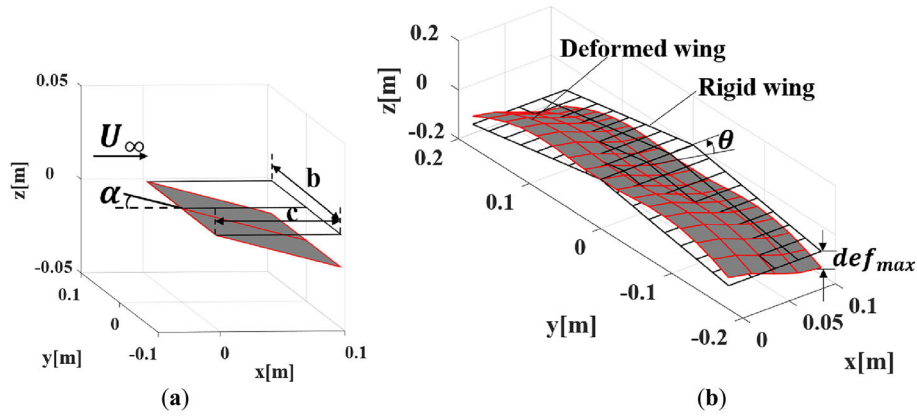


Figure 2. (a) Wing at a specific angle of attack and (b) deformed wing shape. The wing is subjected to both bending ($def_{max} = 0.3b$) and twisting ($\theta = 20^\circ$).

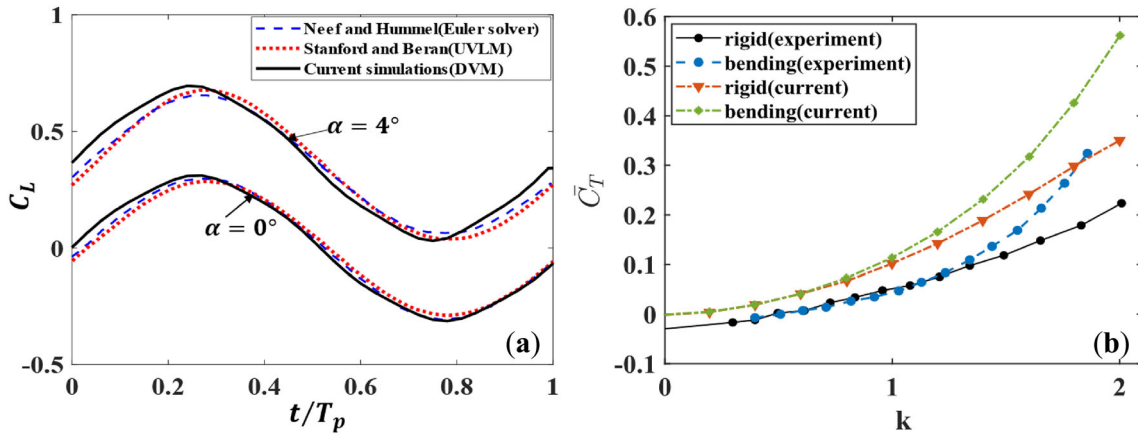


Figure 3. (a) Lift comparison for a flapping wing, showing results from the Euler method [22] and the unsteady vortex lattice method (UVLM) [2]. (b) Thrust comparison for a bending wing flapping motion, with experimental results [28].

varying angle of attack across the wing span and preserving attached flow. Several flying species also exhibit the twisting behaviour during flapping motion [27]. We adopt the method of Wu *et al* [23] to specify the predetermined deformation of the wing, which involves both bending and twisting. The maximum deflection in the bending mode is set to 0.3 times the wing span b , while for the twisting mode, the maximum twisting angle (θ) at the root of the trailing edge is set to 20° . The angular motion during one cycle can be mathematically described using sinusoidal functions, where $\delta(t) = \delta_0 + \delta_1 \sin(\omega t)$, with δ_0 , δ_1 , and ω representing the angular amplitude, mean angle and angular frequency of flapping, respectively. The variable t denotes time in seconds, while T denotes the duration of one flapping cycle.

For the present numerical investigation, we have selected specific parameter values, including a flapping amplitude δ_1 of 30° , an initial inclination δ_0 of -30° , a flapping frequency of 10.5, a reduced frequency ($k = \frac{\pi f c}{U_\infty}$) of 0.3, a chord length (c) of 0.1 m, and a velocity (U_∞) of 11 m/s. The angle of attack (α) is varied within a range of 0° to 25° .

These parameter values have been adopted from prior studies [22, 29, 30]. The time step (Δt) is set to 0.001 s, with a wake vortex being released at a distance of $0.3U_\infty \Delta t$ from the wing tip at each time step [12].

4. Validation

To validate the DVM in capturing the flow dynamics around the flapping wings in forward flight and predicting aerodynamic loads, we analyzed a rectangular wing with an aspect ratio (AR) of 8, a flapping amplitude of $\delta = 15^\circ$, and a reduced frequency of $k = 0.1$. This configuration aligns with prior investigations by Neef and Hummel [22] and Stanford and Beran [2]. Our results show minimal deviation (4–6%) from their data at angles of attack of 0° and 4° (figure 3(a)). One potential source of variation lies in the treatment of shed vortex distance: while our study explicitly defines this parameter as $0.3U_\infty \Delta t$, the earlier works [2, 22] do not specify this value. Despite this difference, the

lift coefficients from our DVM simulations closely match those obtained using the Euler solver [22] and unsteady vortex lattice method (UVLM) [2]. Additionally, figure 3(b) compares thrust coefficients for the rigid and flexible wings across different reduced frequencies. At lower frequencies, our model aligns well with experimental data [28], though a slight overprediction of thrust is observed, probably due to the inviscid assumptions inherent in our approach.

5. Results and discussion

Figure 4 shows the temporal variation of

$$C_L = \frac{L}{\frac{1}{2} \rho U_\infty^2}$$

over two flapping cycles for rigid and deformable wings. Studies [27, 31, 32] indicate that most net lift occurs

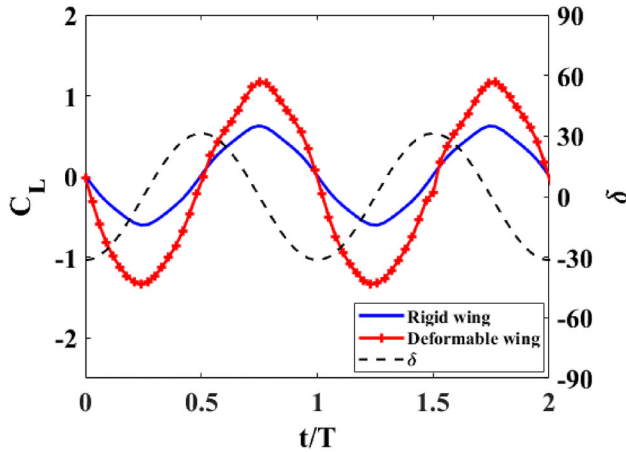


Figure 4. Lift coefficient plot over two cycles at angle of attack ($\alpha = 0^\circ$); $\delta_1 = 30^\circ$; $\delta_0 = 0^\circ$ and $k = 0.3$. In the deformable wing, the peak lift is higher in both strokes, which results in an overall increase in lift compared to the rigid wing.

during the downstroke, with minimal lift during the upstroke. Birds may reduce span, drag, and power use during the upstroke [33]. Introducing a twist in flapping motion significantly alters lift magnitude, with positive lift mainly during the downstroke (figure 4). Higher circulation strength increases lift and thrust (figure 4). For the deformable wing, increased circulation during the downstroke boosts peak lift and thrust. During the upstroke, Lift decreases from $\frac{L}{T_p} = 0$ to $\frac{L}{T_p} = 0.25$ for the rigid wing, then increases positively as the wing moves upward. Peak lift occurs at mid-downstroke ($\frac{0.75t}{T_p}$), consistent with prior studies [2, 22]. The rigid wing generates equal amounts of positive and negative lift, leading to a net zero lift.

The distribution of pressure over the surface of the wing at different positions of the downstroke and upstroke motions of the rigid and deformable wings (a combination of bending and twisting) are highlighted in figure 5(a and b), respectively. The distribution of pressure plays a crucial role in the forces exerted on the body, making it an essential element for optimizing hydrodynamic performance. The pressure is expressed as the pressure coefficient, denoted as

$$C_{p_{ij}} = \frac{P_{ij}}{\frac{1}{2} \rho U_\infty^2},$$

where the pressure on the body is measured as gauge pressure. Following Newton's third law of motion, the force imparted to the fluid behind the wings in the negative x direction (figure 2) produces an equal and opposing force on the body, resulting in the generation of thrust. The net force that propels the object forward, known as the thrust force, arises due to the high-pressure area in front of an object and a low-pressure area behind it (refer figure 5(a and b)). The downstroke motion generates the highest-pressure level on the wings, and a deformable wing experiences higher pressure levels, leading to an increase in lift. In comparison to the rigid wing, the pressure level is higher

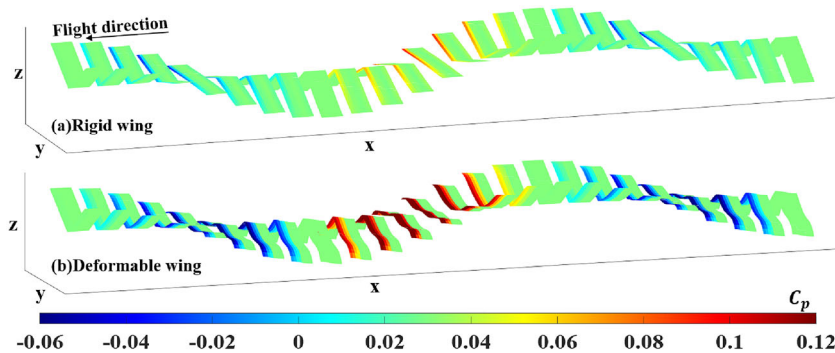


Figure 5. The pressure coefficient distribution over one cycle depicted for the flapping wing with an optimal angle of attack and specific parameters 20: (a) Rigid ($\alpha = 20^\circ$) and (b) deformable wing ($\alpha = 20^\circ$). In the upstroke, there is negative pressure, whereas in the downstroke, the pressure distribution is positive. This indicates that the positive lift is primarily achieved during the downstroke.

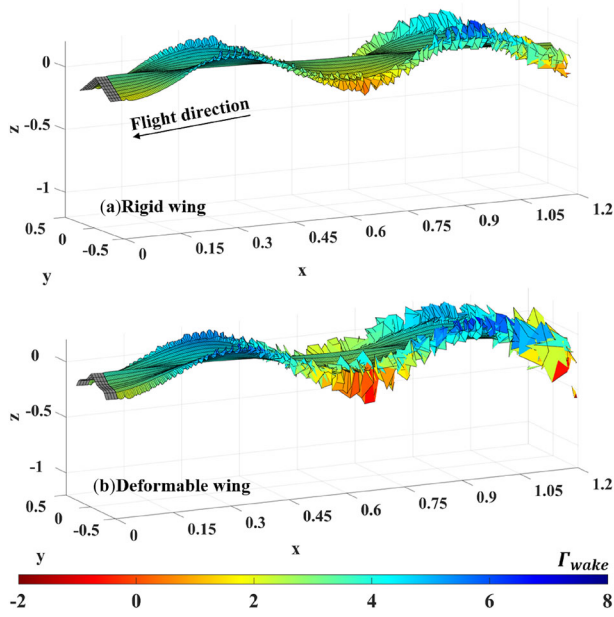


Figure 6. Illustration of wake surfaces following two cycles for a flapping wing at $\delta_1 = 30^\circ$; $\delta_0 = 0^\circ$; $k = 0.3$: (a) Rigid ($\alpha = 0^\circ$) and (b) deformable wing ($\alpha = 0^\circ$). The deformable wing exhibits a more complex flow structure, suggesting enhanced aerodynamic performance compared to the rigid wing.

during both the downstroke and upstroke phases of the deformable wing.

Figure 6(a and b) illustrate the strength of vorticity circulation of the wakes acquired for the rigid and deformable wings, respectively at $\alpha = 0^\circ$. We have observed that the strength of the vorticity circulation in a rigid wing is less than a deformable wing (see figure 6(a and b)). As stated by Kelvin's circulation theorem, the total circulation around a flapping wing and its wake remains unchanged over time

$$\frac{D\Gamma_{\text{wing}}}{Dt} + \frac{D\Gamma_{\text{wake}}}{Dt} = 0.$$

This enhanced vorticity circulation in deformable wings results in a greater overall lift force than that in rigid wings. As the wake's circulation increases, the circulation around the wing also increases in proportion. Kelvin's circulation theorem ensures that any changes in the wake's circulation directly affect the wing's circulation. An increase in wake circulation alters the pressure distribution and flow patterns around the wing, which in turn affects the generation of lift.

Additionally, the wing's lower surface typically experiences a high-pressure zone, while the upper surface exhibits a low-pressure zone. When the wing twists, the high-pressure zone may reconnect abruptly with the wing surface, causing a rapid release of energy into the flow, thereby strengthening the vortex. This results in an enhanced circulation strength in the twisted wing, which significantly impacts the induced velocity field. As a result, there is a marked increase in the lift produced by the wing.

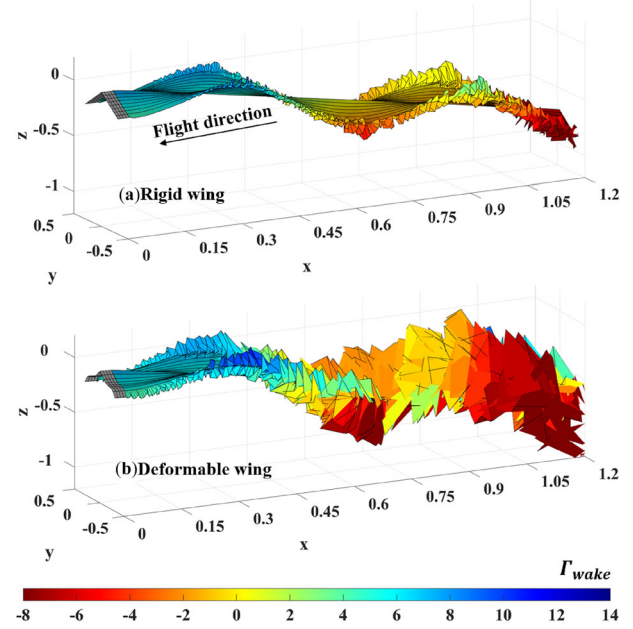


Figure 7. Illustration of wake surfaces following two cycles for a flapping wing at $\delta_1 = 30^\circ$; $\delta_0 = 0^\circ$; $k = 0.3$: (a) Rigid wing ($\alpha = 20^\circ$) and (b) deformable wing ($\alpha = 20^\circ$), showing the influence of flexibility on wake vorticity and downstream structures.

Consequently, it can be concluded that introducing a twist to the wing is the most effective approach for generating lift compared to other methods.

Additionally, we are also examining the influence of the angle of attack on the flapping wing. The angle of attack is a critical parameter that significantly impacts the performance of flapping wings. It represents the angle formed by the wing's chord line and the direction of the incoming airflow. Figure 7 illustrates the wake surface at an angle of attack (α) of 20° . At this high angle of attack, a strong vortex is formed, and the vortex primarily propagates in the downward direction. This downward motion results from the fluid momentum being directed downward as the wing interacts with the incoming flow. Consequently, the wing experiences increased vertical forces, enhancing the overall lift. Notably, the deformable wing generates a stronger vortex than a rigid wing, further improving lift performance. The reason is that a higher angle of attack amplifies the pressure disparity between the lower and upper surfaces of the wing, resulting in an increased vorticity circulation (figure 7, also refer [9]). This increased vorticity circulation generates an augmented lift force. However, if the angle of attack exceeds a certain value, the airflow over the wing may experience flow separation and stall, resulting in reduced lift. This angle of attack, known as the stall angle, should be avoided to achieve optimum aerodynamic performance. Higher stall angle of 20° is observed in deformable wings than in a rigid wing of 15° . Thus,

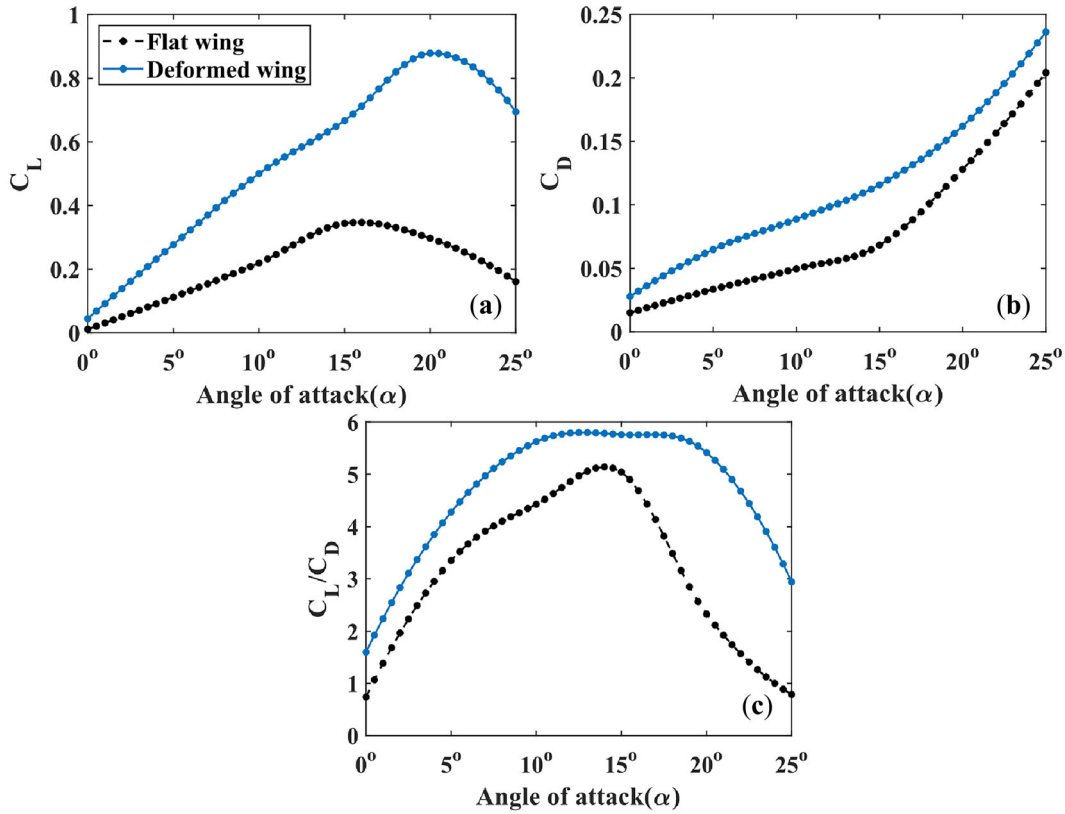


Figure 8. The influence of α on aerodynamic performance C_L , C_D , and C_L/C_D is analyzed based on results from 10 cycles of flapping motion with $\delta_1 = 30^\circ$; $\delta_0 = 0^\circ$; $k = 0.3$: (a) C_L , (b) C_D , and (c) C_L/C_D .

incorporating a deformable wing can help delay the stall angle.

The estimation of lift coefficient C_L , drag coefficient C_D , and lift-to-drag ratio $\frac{C_L}{C_D}$ at several angles of attack α is crucial for safe operation and optimum aerodynamic performance. Figure 8 illustrates how the α affects the lift $C_L = L/(\frac{1}{2}\rho U_\infty^2)$ and drag $C_D = D/(\frac{1}{2}\rho U_\infty^2)$ coefficients and the ratio of lift-to-drag C_L/C_D for both the rigid and deformable wing configurations. Figure 8(a) illustrates the lift coefficient (C_L) trends, showing that the flat wing exhibits a steady increase in lift with rising angles of attack (α) until reaching a peak, followed by a sharp stall. In contrast, the deformed wing shows a more gradual increase in lift and a delayed stall, suggesting improved aerodynamic efficiency and stability at higher angles. The rigid wing achieves its highest lift coefficient at $\alpha = 15^\circ$. In contrast, the deformable wing reaches its peak at $\alpha = 20^\circ$, representing a 55.25% increase compared to the rigid wing's maximum lift coefficient at $\alpha = 15^\circ$. Figure 8(b) depicts the drag coefficient (C_D), with the rigid wing exhibiting lower drag due to reduced turbulent wake and pressure drag. Figure 8(c) presents the lift-to-drag ratio (L/D), highlighting the deformed wing's superior efficiency, particularly at higher angles of attack. Overall, the results emphasize the benefits of wing deformation, including delayed stall and enhanced L/D ratios.

6. Conclusion

For this study, we conducted numerical simulations to analyse the three-dimensional flapping motion of a rectangular wing employing the DVM approach. To ensure the accuracy of our simulations, we verified our DVM code by comparing it with a prior study that examined a rectangular wing in forward motion with an aspect ratio of 8 and Reynolds numbers on the order of 70,000. The primary focus of our work was to investigate the lift and drag performance of the 3D flapping wing. Our analysis focuses on the influence of the angle of attack on the performance of a deformable wing that is subjected to a combination of bending and twisting. Our study reveals that the deformable wing generates greater lift than the rigid wing, but only a marginal increase in drag are observed. Overall, the lift-to-drag ratio proves that deformable wings are more efficient in generating lift than rigid at a moderate range of α , typically between 10 and 20.

In summary, our study illustrates the effectiveness of employing vortex methods to simulate unsteady flapping in flows with moderately low Reynolds numbers. These findings can provide a foundation for enhancing the design of flapping MAVs. In the future, our focus will be on aerostructural optimization. The findings presented above can serve as a viable foundation for shaping the

aerostructure. With the knowledge of the ideal cyclic wing shape and its corresponding pressure distribution, we can integrate these aerodynamic forces into a structural model to calculate passive deformations. The internal configuration of the wing structure can then be fine-tuned until it aligns with the previously mentioned deformations.

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References

- [1] Satish K C, Bret K S, Carlos E S C, and Wei S 2009 Flapping wing CFD/CSD aeroelastic formulation based on a co-rotational shell finite element. In: *50th AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference*. p. 2412
- [2] Stanford B K and Beran P S 2010 Analytical sensitivity analysis of an unsteady vortex-lattice method for flapping wing optimization. *J. Aircraft* 47(2): 647–662
- [3] Chang-kwon K and Wei S 2012 Effects of flexibility on the aerodynamics of a hovering flexible airfoil at Reynolds Number of 100 to 1000. In: *50th AIAA Aerospace Sciences Meeting including the New Horizons Forum and Aerospace Exposition*. p. 1206
- [4] Petricca L, Ohlckers P and Grinde C 2011 Micro- and nano-air vehicles: State of the art. *Int. J. Aerospace Eng.* 2011(1): 214549.
- [5] Shyy W, Berg M and Ljungqvist D 1999 Flapping and flexible wings for biological and micro air vehicles. *Prog. Aerospace Sci.* 35(5): 455–505.
- [6] Ansar S A, Zbikowski R and Knowles K 2006 Non-linear unsteady aerodynamic model for insect-like flapping wings in the hover. Part 1: Methodology and analysis. *Proceedings of the Institution of Mechanical Engineers, Part G: J. Aerospace Eng.* 220(2): 61–83
- [7] Faux D, Thomas O, Cattani E and Grondel S 2018 Two modes resonant combined motion for insect wings kinematics reproduction and lift generation. *Europhys. Lett.* 121(6): 66001
- [8] Jafferis N T, Farrell H E, Karpelson M and Wood R J 2019 Untethered flight of an insect-sized flapping wing microscale aerial vehicle. *Nature* 570(7762): 491–495
- [9] Long L N and Fritz T E 2004 Object-oriented unsteady vortex lattice method for flapping flight. *J. Aircraft* 41(6): 1275–1290
- [10] McCroskey W J 1982 Unsteady airfoils. *Ann. Rev. Fluid Mech.* 14(1): 285–311
- [11] Gordon J L 2006 *Principles of Helicopter Aerodynamics with CD Extra*. Cambridge University Press
- [12] Katz J and Plotkin A 2001 *Low-speed Aerodynamics*, vol 13. Cambridge University Press
- [13] Kumar R, Padhee S S and Samanta D 2024 Numerical simulations of bio-inspired approaches to enhance underwater swimming efficiency. *Phys. Fluids*. 36(5).
- [14] Kumar R, Padhee S S and Samanta D 2023 *Aerodynamic performance and flow mechanism of 3D flapping wing using discrete vortex method*. [arXiv:2309.14832](https://arxiv.org/abs/2309.14832)
- [15] Kumar R, Padhee S S and Samanta D 2024 Aerodynamic performance and flow mechanism of 3D flapping wing using discrete vortex method. *J. Fluids Struct.* 127: 104128
- [16] Christopher C, Richard S, Phil B and Gregory P 2009 The physics of an optimized flapping wing micro air vehicle. In: *47th AIAA Aerospace Sciences Meeting Including The New Horizons Forum and Aerospace Exposition*. p. 801
- [17] Jeff E, Chengjie W and Michael OI 2009 A computational study of a canonical pitch-up, pitch-down wing maneuver. In: *39th AIAA Fluid Dynamics Conference*. p. 3687
- [18] Robertson D, Reich G and Joo J 2010 Vortex particle aerodynamic modelling of perching manoeuvres with micro air vehicles. In: *51st AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference*. p. 2825. <https://arc.aiaa.org/doi/pdf/10.2514/6.2010-2825>
- [19] Monaghan J J and Humble R J 1993 Vortex particle methods for periodic channel flow. *J. Comput. Phys.* 107(1): 152–159
- [20] Xu C and Yeung W W H 1996 Discrete vortex method for airfoil with unsteady separated flow. *J. Aircraft* 33(6): 1208–1210
- [21] Nikhil M and Fred H 2006 Gust response of separated shear layers using discrete vortex methods. In *44th AIAA Aerospace Sciences Meeting and Exhibit*. p. 1298
- [22] Neef M F and Hummel D 2001 Euler solutions for a finite-span flapping wing. *Fixed and flapping wing aerodynamics for micro air vehicle applications*. https://books.google.co.in/books?hl=en&lr=&id=q6oX4o8jSuAC&oi=fnd&pg=PA429&dq=Neef+M+F+and+Hummel+D+2001+Euler+solutions+for+a+finite+523+span+flapping+wing.+Fixed+and+flapping+wing+aerodynamics+524+for+micro+air+vehicle+applications,+pp+429%E2%80%9393451&ots=DqVO39buhg&sig=E_1PuhUc6QDOEp6l1rHdRn_4dww&redir_esc=y#v=onepage&q&f=false
- [23] Wu P, Stanford B K, Sällström E, Ukeiley L and Ifju P G 2011 Structural dynamics and aerodynamics measurements of biologically inspired flexible flapping wings. *Bioinsp. Biomimet.* 6(1): 16009
- [24] Ghommam M, Collier N, Niemi A H and Calo V M 2014 On the shape optimization of flapping wings and their performance analysis. *Aerospace Sci. Technol.* 32(1): 274–292
- [25] Devranjan S, Jalikop S V and Sreenivas K R 2013 Study of flapping flight using discrete vortex method based simulations. *Int. J. Mod. Phys. C* 24(12): 1–8
- [26] Soumen C and Devranjan S 2021 Numerical simulation of a one-dimensional flexible filament mimicking anguilliform mode of swimming using discrete vortex method. *Phys. Rev. Fluids* 6(3): 33102
- [27] Norberg U M 1990 Morphological adaptations for flight. In: *Vertebrate flight. Zoophysiology*, Vol 27. Springer, Berlin, Heidelberg. <https://doi.org/10.1007/978-3-642-83848-4-11>

- [28] Heathcote S, Wang Z and Gursul I 2008 Effect of spanwise flexibility on flapping wing propulsion. *J. Fluids Struct.* 24(2): 183–199
- [29] Robert L H 1951 Experimental aerodynamic derivatives of a sinusoidally oscillating airfoil in two-dimensional flow. Vol. 2465. National Advisory Committee for Aeronautics
- [30] Díaz-Arriba D, Jardin T, Gourdain N, Pons F and David L 2022 Experiments and numerical simulations on hovering three-dimensional flexible flapping wings. *Bioinspir. Biometr.* 17(6) 065006.
- [31] Yongliang Y and Ziwu G 2015 Learning from bat: Aerodynamics of actively morphing wing. *Theor. Appl. Mech. Lett.* 5(1): 13–15
- [32] Chen S, Li H, Guo S, Tong M and Ji B 2018 Unsteady aerodynamic model of flexible flapping wing. *Aerospace Sci. Technol.* 80: 354–367
- [33] Pennycuik C J 1968 Power requirements for horizontal flight in the pigeon *Columba livia*. *J. Exp. Biol.* 49(3): 527–555

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